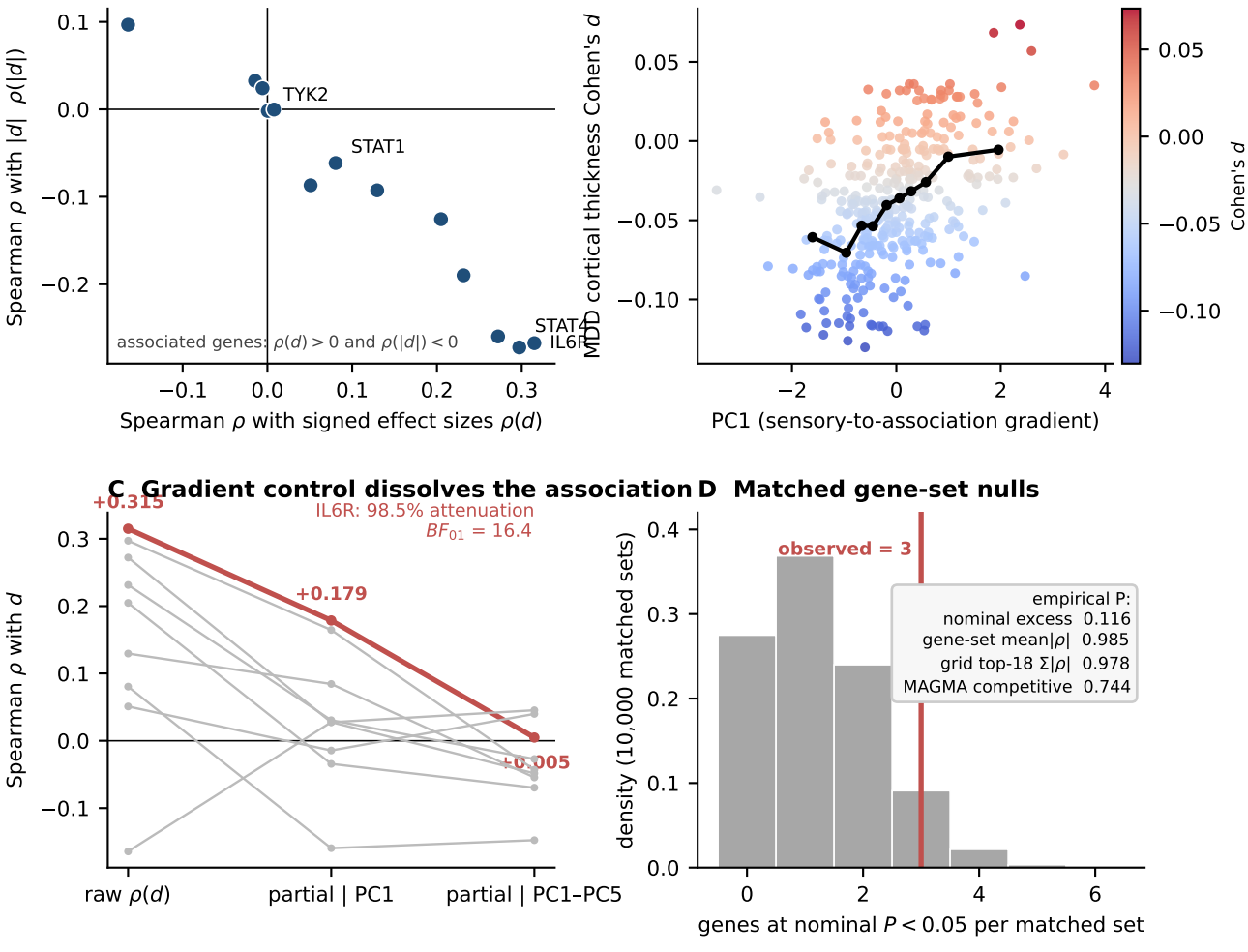

A Sign structure of 13 genes
Spearman ρ with |d| ρ(|d|)
Spearman ρ with signed effect sizes ρ(d)
TYK2
STAT1
STAT4
IL6R
associated genes: ρ(d) > 0 and ρ(|d|) < 0
B Gradient prior: ρ = 0.491, P = 6.1e-22
MDD cortical thickness Cohen's d
PC1 (sensory-to-association gradient)
Cohen's d
C Gradient control dissolves the association
Spearman ρ with d
+0.315
+0.179
+0.005
IL6R: 98.5% attenuation
BF01 = 16.4
raw ρ(d)
partial | PC1
partial | PC1–PC5
D Matched gene-set nulls
density (10,000 matched sets)
observed = 3
empirical P:
nominal excess 0.116
gene-set mean|ρ| 0.985
grid top-18 Σ|ρ| 0.978
MAGMA competitive 0.744
genes at nominal P < 0.05 per matched set